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X Image Model - Teachable Machine

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X teachable machine - بحث

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📁 sample_data

📄 keras_model.h5

📄 labels.txt

📄 tree.jpg

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```
from keras.models import load_model # TensorFlow is required for Keras to work
from PIL import Image, ImageOps # Install pillow instead of PIL
import numpy as np
import tf_keras as tk

# Disable scientific notation for clarity
np.set_printoptions(suppress=True)

# Load the model
model = tk.models.load_model("keras_model.h5", compile=False)

# Load the labels
class_names = open("labels.txt", "r").readlines()

# Create the array of the right shape to feed into the keras model
# The 'length' or number of images you can put into the array is
# determined by the first position in the shape tuple, in this case 1
data = np.ndarray(shape=(1, 224, 224, 3), dtype=np.float32)

# Replace this with the path to your image
image = Image.open("tree.jpg").convert("RGB")

# resizing the image to be at least 224x224 and then cropping from the center
size = (224, 224)
```

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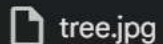
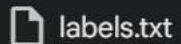
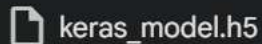
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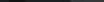
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▶  sample_data

keras_model.h5

 labels.txt tree.jpg



```
[5] # turn the image into a numpy array
image_array = np.asarray(image)
```

```
# turn the image into a numpy array
image_array = np.asarray(image)

# Normalize the image
normalized_image_array = (image_array.astype(np.float32) / 127.5) - 1

# Load the image into the array
data[0] = normalized_image_array

# Predicts the model
prediction = model.predict(data)
index = np.argmax(prediction)
class_name = class_names[index]
confidence_score = prediction[0][index]

# Print prediction and confidence score
print("Class:", class_name[2:], end="")
print("Confidence Score:", confidence_score)
```

```

v ... 1/1 [=====] - 1s 1s/step
    Class: oak tree
    Confidence Score: 0.99896646

```

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